Flight Right

Trajectory Prediction — Results and Next Steps

Status report · 1 September 2026

This report explains, in plain language, what the aircraft-position prediction model currently does, how well it works, why it behaves the way it does, and what has to change. No aviation background is assumed.

> **The one-line summary**
> The AI model is currently worse than simple arithmetic at short-range prediction, and only slightly better at long range. The cause is understood and fixable. However, the 95% accuracy target is achievable at 1 and 3 minutes but **not** at 5 or 10 minutes — not because of the model, but because the necessary information does not exist in the data we collect.

1. What the system is trying to do

We collect the live position of every aircraft flying near Mumbai, roughly every 10 seconds. The goal is to look at an aircraft's recent movement and predict where it will be **1, 3, 5 and 10 minutes into the future**. If we can do that, we can spot two aircraft heading for the same piece of sky before it happens, and warn the controller.

Everything in this report is about that prediction step.

The data we have

18 collection sessions of 90 minutes each, gathered between 24 August and 1 September. That is about **322,000 aircraft position readings** covering **1,164 different aircraft**. From these we built **63,939 training examples**. Each example is 10 consecutive position readings of one aircraft, paired with where that aircraft actually turned out to be 1, 3, 5 and 10 minutes later.

We split those examples into three groups: the model learns from 45,377, we tune it using 10,406, and we test it on 8,156 it has never seen. Crucially, the test examples come from **a later day** than the training ones — so the test is genuinely 'predict the future', not 'recall the past'.

2. The simple method we have to beat

Before building anything clever, we built the dumbest possible predictor, called the **constant-velocity baseline**.

> **How it works — the car analogy**
> Imagine you see a car on a highway heading north at 100 km/h. Where will it be in 10 minutes? You would guess: about 17 km further north. You assume it keeps going the same direction at the same speed.
>
> That is exactly what this method does with aircraft. No learning, no training, no AI — just multiplication.

It sounds too simple to be useful. It is not. Most aircraft are cruising in a straight line at constant speed, so the guess is often very good. Here is a real example from our data — an aircraft cruising at 477 knots:

Predicting ahead by	How far off the simple guess was

1 minute	0.11 nautical miles
3 minutes	0.57 nautical miles
5 minutes	0.78 nautical miles
10 minutes	1.38 nautical miles

That aircraft travelled 79 nautical miles in those ten minutes, and the naive guess was off by 1.38 — an error of under 2%, using nothing but arithmetic.

> **Why this matters for the project**
> This is the bar the AI has to clear. A neural network with 400,000 internal settings, a GPU and hours of training has to justify itself against multiplication. If it cannot beat this, the complexity is buying nothing. This is why the project rules require building the simple method first and keeping it permanently as a safety net.

3. What the AI model actually achieved

We trained the AI model (a Bi-LSTM neural network) on the GPU and tested it against the simple method on identical data. 'Within 5 miles' is the useful measure — that is roughly the spacing controllers keep between aircraft.

Predicting ahead	Simple method within 5 miles	AI model within 5 miles	Verdict
1 minute	100.0%	60.5%	AI much worse
3 minutes	94.4%	57.1%	AI much worse
5 minutes	83.9%	54.3%	AI worse
10 minutes	55.4%	46.8%	AI slightly worse

Why the AI model failed

Look at how the AI model's error changes as we predict further ahead, compared with the simple method:

Predicting ahead	1 min	3 min	5 min	10 min
Simple method — average error	0.6 mi	1.6 mi	3.4 mi	10.4 mi
AI model — average error	5.2 mi	5.5 mi	6.0 mi	7.5 mi

The simple method's error grows steadily the further ahead you look — exactly what you would expect. The AI model's error barely changes: it is almost as wrong at 1 minute as at 10.

> **What that means in plain terms**
> The model stopped tracking individual aircraft and started guessing roughly where planes generally are. It learned to answer 'somewhere over the Mumbai region' rather than 'this particular aircraft, here'. It is playing it safe by predicting the average.

The cause is a specific technical mistake, and it is fixable. During training the model is scored on all four time horizons at once, added together. The 10-minute distances are about 17 times larger than the 1-minute ones, so they completely dominate the score. The model learns that getting 10 minutes roughly right matters, and that 1 minute is not worth bothering with.

4. What makes an aircraft hard to predict

We measured this rather than guessing. Two things break the prediction, and both are intuitive.

Turning

If an aircraft is flying straight, guessing where it goes is easy. If it is turning, it is not.

How much the aircraft turned	Number of cases	Predictions within 5 miles
Almost straight (under 1 degree)	36,625	58%
Gentle turn (10–30 degrees)	2,369	36%
Sharp turn (over 30 degrees)	1,565	10%

Climbing or descending

A level aircraft is far easier to predict than one changing altitude — because a climbing or descending aircraft is usually also being given turn instructions by a controller.

What the aircraft was doing	Number of cases	Predictions within 5 miles
Flying level	32,572	70%
Climbing or descending gently	3,676	29%
Climbing or descending fast	7,969	10%

> **The headline finding**
> A climbing or descending aircraft is about **7 times harder** to predict than a level one. A sharply turning aircraft is about **6 times harder** than one flying straight. Roughly a third of all aircraft are doing one of these at any moment, and they account for almost all of the total error.

5. Which factors matter most

We ran a statistical test to find which pieces of information best predict whether a prediction will be accurate. The test explains 66% of the variation, which means the errors are largely predictable rather than random — and that is the room the AI has to improve in.

Rank	Factor	Importance	What it tells us in plain terms
1	Altitude	29%	High up means cruising, which means flying straight
2	Distance from Mumbai airport	29%	Near the airport means turning, queueing, descending
3	Speed	17%	Fast means cruising; slow means approaching or departing
4	How sharply it is turning	7%	Direct measure of the biggest failure mode
5	Rate of climb or descent	5%	The second biggest failure mode
6	Total turn so far	3%	Catches long, steady turns

> **The insight**
> The top three factors carry 75% of the signal, and they are all really answering one question: **is this aircraft cruising, or is it near the airport doing something complicated?** Cruising aircraft are easy. Aircraft near the airport are hard. The model does not need to understand aviation — it needs to know which of those two situations it is looking at.

What we should feed the model

At present the model is given six raw numbers: position, altitude, speed, direction and rate of climb. **None of them describe change.** The model is never told 'this aircraft is turning' — it has to work that out itself from a sequence of compass readings. We should calculate it and hand it over directly. Six additions, in order of expected value:

New information to provide	Why it helps
Distance to Mumbai airport	Tied for the most important factor; currently the model has to infer it
How fast it is turning	The strongest single warning sign of an unpredictable aircraft
Total turn over the window	Catches long steady turns that a snapshot misses
Whether it is speeding up	Indicates a change of flight phase
Change in climb rate	Distinguishes levelling off from continuing to climb
Direction to the airport	Separates arriving from departing from passing overhead

6. The 95% accuracy target — an honest answer

The target set was above 95% accuracy at every time horizon. Based on the measurements, **this is achievable at 1 and 3 minutes, but not at 5 or 10 minutes** — and the reason is not the model.

To test the ceiling, we took the easiest possible cases: aircraft flying level, straight and high up, the most predictable 62% of all traffic, and measured how well even the simple method does on those alone.

Predicting ahead	All aircraft within 5 miles	Easiest aircraft only within 5 miles	Distance needed to capture 95%
1 minute	100%	100%	1.5 miles
3 minutes	94%	99%	5.5 miles
5 minutes	84%	93%	13 miles
10 minutes	55%	70%	37 miles

Why 10 minutes cannot reach 95% — the road-trip analogy
In 10 minutes an aircraft flies about 74 miles. Landing within 5 miles of the truth means predicting its position to within 7% of the distance it covers.

It is like watching a car on a motorway and being asked exactly where it will be in an hour. You can see its speed and direction, but you cannot see its **destination**. Whether it takes the next exit is decided by the driver, not by anything visible from outside.

Aircraft are the same. Where a plane will be in 10 minutes depends on its filed flight plan and the instructions air traffic control gives it. Neither is in the data we receive. Even for the easiest, straightest, highest aircraft, the simple method only reaches 70% — and that is a limit of the information, not of the software.

Targets we can actually defend

Predicting ahead	Realistic target	Status
1 minute	95% within 2 miles	Already achieved by the simple method
3 minutes	95% within 5 miles	Reachable — currently at 94%
5 minutes	95% within 10 miles	Reachable
10 minutes	70–75% within 5 miles	Realistic ceiling without flight plan data

This is a stronger result for the project than a number we cannot defend. Saying '95% accurate at 3 minutes, 70% at 10 minutes, limited by the absence of flight plan data' shows we understand the problem. Claiming 95% everywhere would invite the immediate question of how, and there is no good answer.

7. What to do next

#	Action	Why it matters
1	Fix how the model is scored during training	**Nothing else matters until this is done.** Score each time horizon separately so short-range predictions stop being ignored. Also predict *how far the aircraft moves* rather than its absolute map position.
2	Add the six new pieces of information	Especially distance to the airport and turn rate — the two strongest factors, neither of which the model currently receives.
3	Retrain and re-measure	Expect 3-minute predictions to pass 95%, and 10-minute to settle around 70%.
4	Fix the data collection scheduling	Roughly a third of collection sessions are being killed when the laptop sleeps or locks. Fixing this is a settings change, and more data helps every number above.
5	Keep the simple method permanently	It beats the AI at short range and always will. The sensible design uses the simple method for 1–5 minutes and the AI for 10 minutes, with the simple method as a fallback if the AI ever misbehaves.

How to summarise this to someone else
"We built a simple physics-based predictor first, then an AI model, and measured both honestly. The AI currently loses at short range because of a scoring flaw we have identified and can fix. We also proved which factors actually make aircraft hard to predict — turning and changing altitude — and confirmed that predicting 10 minutes ahead to 95% accuracy is impossible without flight plan data we do not have access to. So we are targeting 95% at 3 minutes and about 70% at 10 minutes, and keeping the simple method as a permanent safety net."